

A History of Conversational AI

From the Turing Test to the LLM Era: A Relay Race of Relocating Bottlenecks

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Abstract

Conversational machines have travelled from a 200-line pattern substituter that unsettled its own author in 1966 to systems that hundreds of millions of people now address daily, yet the standard telling—ever-smarter algorithms—explains neither why some capabilities arrived decades after the idea behind them nor why today’s frontier is partitioned by who can buy accelerators. We survey the full 1950–2026 arc of text-first open-domain and task-oriented conversational systems and argue a single contestable thesis: the history is not a story of cumulative intelligence but a relay race of relocating bottlenecks, in which each era won by removing whichever resource was scarce—hand-authoring labor, then labeled data, then reusable representations, then raw compute and web text, then human-preference feedback—and the binding constraint has now migrated out of the model architecture entirely, into training-data exhaustion and geopolitical access to compute. Reducing every era to four things—its paradigm, its binding constraint, the unlock that relaxed it, and the residue it left unsolved—we organize the sweep around its transitions rather than its calendar, anchor an eighteen-system cross-cutting matrix to the frame, and trace an evaluation thread that saturates in lockstep with each constraint’s removal. We steelman the opposing view, that this is one continuous arc of genuine understanding driven by modeling ideas, and reject it on the evidence of dormancy lags and the export-control natural experiment. The frame yields falsifiable predictions, each with a stated kill-criterion: if the next frontier leap comes from a compute-poor lab on a modeling idea alone, the thesis is wrong.

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1 Introduction

In 1966, Joseph Weizenbaum wrote a program of a few hundred lines that matched patterns in a user’s typing and reflected them back as questions. ELIZA had no model of meaning, no memory beyond the current sentence, and no knowledge of the world; its author built it partly to demonstrate how *little* was needed to produce the appearance of understanding. He was then disturbed to find his own secretary asking him to leave the room so she could talk to it in private. Sixty years later, conversational systems descended from a very different lineage answer hundreds of millions of people every day, write working code, and pass professional examinations. The obvious story connecting these two endpoints is that the machines simply got smarter. This survey argues that the obvious story is wrong in a specific and consequential way.

The wrongness is not that progress is illusory—it is plainly real—but that “got smarter” misidentifies the engine, and misidentifying the engine has costs. A field that believes it advances by accumulating better algorithms will look for its next breakthrough in architectures and training objectives, will read the geographic distribution of capability as a map of where the best ideas are, and will treat data and serving economics as engineering footnotes beneath the real science. We will argue that each of these readings is mistaken, that the history supports a different and more predictive account, and that the difference matters most precisely now, when the resources the field actually runs on have become scarce and contested. The remainder of this introduction states why the question is forced today, what account we propose in its place, and how the rest of the survey earns it.

1.1 Why Now?

The question is forced now because the present frontier has made the usual explanation visibly incomplete. If conversational progress were a smooth accumulation of better algorithms, two facts would be puzzling. First, several of the decisive ideas—neural language models, learning from human preferences, and the inference-time elicitation of multi-step reasoning—existed years before they produced any conversational payoff, and sat inert until something other than a new idea changed. Second, the current distribution of frontier capability does not track where the best modeling ideas are; it tracks who can obtain advanced accelerators, to the point that two governments have chosen to contest conversational AI by restricting *hardware* rather than ideas, which diffuse across borders in days. A history that cannot explain late arrivals and a compute-drawn frontier map is missing its causal engine.

We propose that engine. The history of conversational AI is not a story of steadily smarter algorithms but a relay race of relocating bottlenecks: each era won not by becoming globally more intelligent but by removing whichever single resource was scarce—hand-authoring labor, then labeled data, then usable representations, then raw compute and web-scale text, then human-preference feedback—and the binding constraint has now migrated out of the model architecture entirely, into data exhaustion and geopolitical access to compute. Progress is real; its cause is constraint-relocation, not cumulative understanding. To make this legible across a seventy-six-year span we reduce every system and era to four things: its *paradigm* (the dominant mechanism for deciding what to say), its *binding constraint* (the one scarce resource that capped it), its *unlock* (the concrete artifact that relaxed the prior constraint), and its *residue* (what persisted unsolved—above all the gap between perceived and actual understanding that ELIZA exposed). We organize the survey around the *transitions* between paradigms rather than the calendar, because the transitions are where the thesis lives.

This survey covers 252 sources spanning 1950–2026—peer-reviewed papers, lab technical reports, and model cards—organized around five argument steps and their three principal objections, and it anchors an eighteen-system cross-cutting comparison matrix (Table 1) to the binding-constraint frame. We treat contested historical claims as claims to be traced to primary sources rather than repeated from folklore, and we confirm the run-date frontier empirically rather than on faith.

1.2 Relationship to Existing Surveys

Existing overviews are largely catalogues of methods or systems; this survey is an argument. We defer the detailed comparison to §2, but in brief: where recent LLM surveys [Zhao et al., 2026] and evaluation surveys [Chang et al., 2024] document *what* the current models and metrics are, we argue *why* the transitions happened, across a longer arc (from 1950) and under a single causal thesis, and we subordinate the agentic-systems literature [Wang et al., 2024] to that frame rather than treating it as the subject.

1.3 Contributions

We make four contributions.

1. **Binding-Constraint Frame.** We formalize a four-part lens (paradigm, constraint, unlock, residue) and use it to read the entire 1950–2026 sweep as constraint-relocation, stating the frame’s falsifiable content so it cannot collapse into hindsight-fitting (§2).

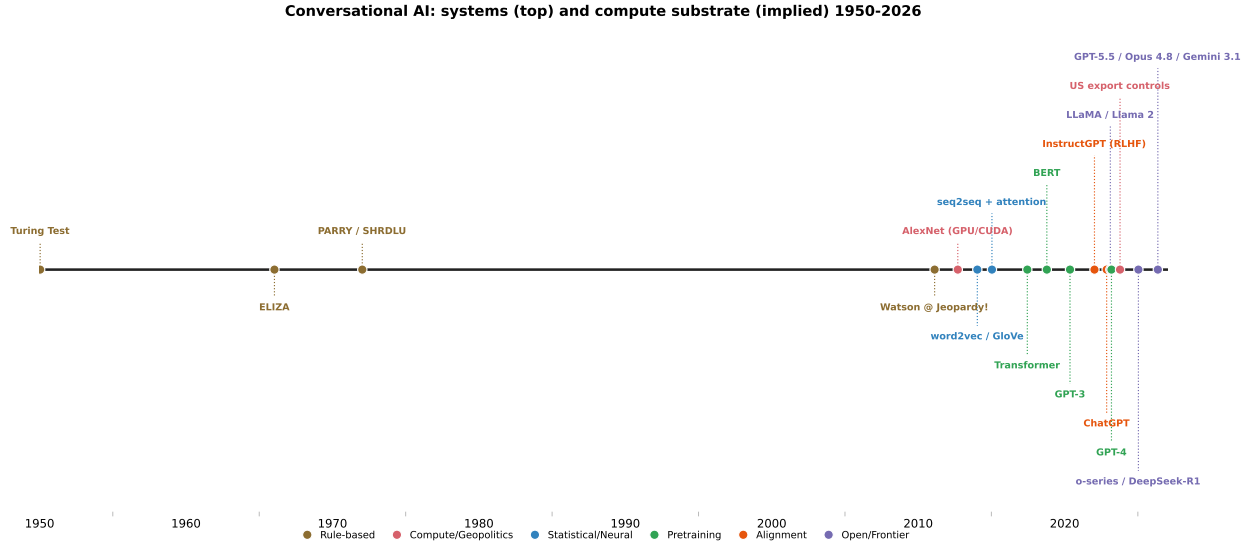


Figure 1: Trajectory of conversational AI, 1950–2026. Named flagship systems and milestones sit on a single time axis, colored by paradigm; the compute and geopolitics milestones (AlexNet/CUDA, the export controls) are interleaved to show that the substrate steps gate the conversational ones. The run-date frontier (GPT-5.5, Claude Opus 4.8, Gemini 3.1 Pro, and the export-constrained Chinese frontier) is the final, fully developed era rather than a coda.

2. **The Constraint-Swap Chain.** We show that the rule-based, statistical, pretraining, and alignment transitions each map onto a single constraint swap, and we assemble an eighteen-system matrix that places every milestone from ELIZA to the run-date frontier on one comparison frame (§3).
3. **Substrate-Primacy Evidence.** We marshal dormancy-lag evidence and the export-control natural experiment to argue, against the naive “models got smarter” reading, that the binding inputs were disproportionately substrate rather than conversational modeling ideas (§4).
4. **Falsifiable Forecasts.** We pair an open-problems analysis with forward predictions, each carrying an explicit kill-criterion tied to the binding-constraint frame (§6).

Section 2 develops the frame and situates the survey against prior overviews. Section 3 walks the constraint-swap chain and presents the cross-cutting matrix. Section 4 argues substrate-primacy from the idea-to-payoff lag; Section 5 extends it to the compute-and-geopolitics layer and the run-date frontier. Section 6 reads the history of evaluation as a sequence of saturating proxies. The final sections state the open problems, the falsifiable directions, and the verdict. Figure 1 previews the whole trajectory on a single axis.

2 Background: The Binding-Constraint Lens

We claim that each era of conversational AI is defined by exactly one binding constraint whose removal—not generic improvement—triggered the transition to the next, and that reducing every era to four things (paradigm, constraint, unlock, residue) makes the whole 1950–2026 sweep legible as constraint-relocation rather than as cumulative intelligence.

The natural and serious objection is that this is too tidy. Seventy-six years of work by thousands of researchers across symbolic AI, statistics, linguistics, and systems engineering did not advance along a single scarce axis at a time; compute, data, algorithms, and engineering practice all improved together and fed back on one another. On this reading, naming one “binding constraint” per era is a narrative imposed after the fact, and the honest description is continuous, multi-causal progress in which ELIZA and GPT-4 differ in degree rather than in what was scarce. A frame that can label any historical sequence with a post-hoc bottleneck explains nothing, because it forbids nothing.

We take the objection seriously and answer it with a definition that carries falsifiable content. A constraint is *binding* in an era when its absence demonstrably blocked the era’s characteristic capability *while the other inputs were already present*—the test is counterfactual leverage, not mere correlation. This is not a frictionless fit: it makes commitments that can fail. It predicts, for instance, that an idea whose enabling resource has not yet arrived will lie dormant regardless of its quality, and that relaxing the named constraint—not supplying a cleverer algorithm—will trigger the next jump. Both claims are checkable, and §4 shows they hold at the points where the rival “better ideas won” story would predict otherwise.

The distinction between counterfactual leverage and correlation is what gives the lens its teeth, and it is worth stating as a procedure rather than a slogan. To license the label *binding* for a given era we ask three questions in order. First, was the resource actually scarce while the era’s characteristic capability was blocked—not scarce in some abstract sense, but scarce at the precise margin where adding more of it would have moved capability? Second, were the *other* inputs already in hand, so that the era was waiting on this one resource and not on a still-missing idea or a still-missing dataset? Third, did capability in fact jump when, and only when, that resource arrived—rather than drifting upward smoothly as every input improved together? Only when all three hold do we name the constraint binding; where they do not, we say the joint is multi-causal and decline to force a single label. This is a deliberately demanding test, and it is the reason the frame can be wrong: an era in which capability climbed steadily while no single resource was decisive, or in which a new idea unlocked progress with every other input already abundant, would not fit, and we would have to report the misfit rather than relabel it.

The four-part reduction operationalizes the lens by forcing every era and every system onto the same four questions, which is what makes a seventy-six-year sweep comparable at all. The *paradigm* is the mechanism for deciding what to say, running from pattern substitution through knowledge engineering, information retrieval, neural sequence transduction, large-scale pretraining, preference alignment, and now tool-augmented reasoning; naming it fixes *how* a system generates an utterance, independent of how well it does so. The *binding constraint* is the one scarce resource whose removal—by the test above—triggered the move to the next paradigm. The *unlock* is the concrete artifact, almost always datable, that relaxed the prior constraint: not a diffuse “advance” but a specific corpus, accelerator, pipeline, or architecture one can point to. And the *residue* is what no era resolved: the gap between perceived and actual understanding that Weizenbaum’s users displayed and that recurs, scaled up, in every subsequent system [Shum et al., 2018]. Tracking the residue as a fifth, unchanging column is what keeps the survey honest about progress—each era relaxed a constraint and widened the surface of the residue at the same time, so the trajectory is one of real capability gains layered over a durable, never-closed gap. That residue is why the “understanding” debate is itself a recurring object of study—Searle’s Chinese Room [Moural, 2003] restated the worry for symbolic systems, and its descendants restate it for each new paradigm. Even the field’s high-water marks of apparent comprehension, such as the question-answering system that won at *Jeopardy!* [Ferrucci et al., 2010], were engineered around the *appearance* of understanding

on a bounded task rather than around closing the residue. Recent comprehensive overviews of large language models confirm how quickly the paradigm-of-record now turns over [Zhao et al., 2026].

2.1 The seven paradigms at a glance

The eras we will traverse are: rule-based pattern substitution (ELIZA, PARRY, SHRDLU); knowledge engineering and competition-driven bots (AIML/A.L.I.C.E., Watson); statistical and distributed-representation methods (word2vec, GloVe, seq2seq with attention); large-scale pretraining (the Transformer, BERT, the GPT line); preference alignment (RLHF, InstructGPT, ChatGPT, LaMDA, Constitutional AI); the open-and-closed model explosion (Llama, Mistral, Gemini, and the Chinese frontier); and the reasoning/agent turn that defines the run-date frontier—each system cited in the section where we examine it. Each, we will argue, is a response to the residue of scarcity left by its predecessor.

The list is not a chronology to be recited but a sequence of *handoffs*, and the survey’s claim is about the handoffs rather than the stations. Read forward, each paradigm inherits the residue its predecessor could not absorb and converts the predecessor’s scarce resource into an abundant one: knowledge engineering industrialized the hand-authoring that rule-based systems did by hand, only to make labeled coverage the new ceiling; statistical methods replaced authored rules with estimated parameters, only to make reusable representations scarce; pretraining made representations cheap, only to make compute and web text the binding inputs; and alignment made raw capability usable, only to make human-preference data the bottleneck. The pattern is mechanical enough that the reader can predict the shape of each transition before seeing its details: the resource that one era spends lavishly is the resource the next era has learned to manufacture. What the list cannot show—and what the rest of the survey supplies—is the evidence that these handoffs were driven by the arrival of a resource rather than by the arrival of a better idea; that is the work of §3 and §4.

2.2 The recurring residue: the ELIZA effect across eras

The single thread that survives every transition is the ELIZA effect: users attribute understanding in proportion to fluency, not in proportion to demonstrated robustness. We treat this not as an anecdote from 1966 but as a measurable, recurring gap—between what a system can reliably do and what its users believe it can do—that each paradigm widened the surface of rather than closed.

The reason the residue is structural rather than incidental is that every constraint the field has relaxed so far has been a constraint on *fluency or coverage*, never on *grounding*. Removing the authoring bottleneck let systems say more things; removing the data bottleneck let them say them in more domains; removing the compute bottleneck let them say them fluently across the open web; and removing the preference-feedback bottleneck let them say them in a register users find helpful and trustworthy. Each of these enlarges the set of utterances a system can produce convincingly without touching the question of whether the system can tell when it is wrong. So the very resources that drive the relay race also widen the residue: a more fluent, more helpful, more broadly competent system is, other things equal, one whose confident errors are *harder* for a non-expert to catch, not easier. The alignment era makes this explicit—optimizing for human approval is optimizing for the perceived side of the gap [Huang et al., 2024]—and the evaluation history of §6 shows every measurement regime inheriting the same blind spot. We therefore return to the residue twice more: as the recurring failure mode in the matrix of §3, and as the survey’s first open problem, where we argue it is not merely unsolved but, at the frontier, essentially unmeasured.

2.3 Relationship to existing surveys

This survey is deliberately not a catalogue of systems or methods, which distinguishes it from the adjacent literature. Zhao et al. [2026] provide a comprehensive technical reference on large language model architectures, training, and capabilities; we cover a longer historical arc (from 1950) and advance a causal thesis about *why* the transitions happened, where they describe *what* the current models are. Chang et al. [2024] survey evaluation methodology for LLMs as a standing field; we instead treat evaluation historically, as a sequence of proxies that each saturated when its era’s binding constraint was removed (§6), rather than as a toolkit to be catalogued. Raiaan et al. [2024] organize LLMs by architecture, application, and taxonomy; our organizing axis is the binding constraint, not the architecture, and our scope reaches back through the rule-based and statistical eras they omit. Wang et al. [2024] survey LLM-based autonomous agents as a contemporary design space; we treat the agentic turn only as the most recent paradigm in a seventy-six-year trajectory and subordinate it to the constraint frame. Roumeliotis and Tselikas [2023] offer an early review of ChatGPT and OpenAI models; we situate that moment as one constraint-swap (the arrival of preference-feedback pipelines) within a longer relay, and we confirm the run-date frontier empirically rather than freezing at a 2023 snapshot.

We concede that the boundaries between eras are not crisp and that constraints genuinely co-move; the representations and compute of the late 2010s arrived nearly together, and some unlocks are part-algorithm and part-pipeline. Where the swap is genuinely simultaneous we say so and mark the relevant matrix cells as shared rather than forcing a single cause. The claim is about which input was binding *at the margin*, not about clean dates or monocausal history.

With the lens defined and its falsifiable content stated, we can now walk the chain of constraint swaps that carried the field from hand-authored rules to human-preference feedback—and show that the chain is visible only from the whole sweep, not from any single paper.

3 The Constraint-Swap Chain: Hand-Authoring to Human Feedback

The rule-based, statistical, pretraining, and alignment transitions each map onto a constraint swap—hand-authoring labor, then labeled in-domain data, then reusable representations, then raw compute and web-scale text, then human-preference feedback—and each era’s unlock is precisely the artifact that relaxed the prior era’s scarce resource. The thesis names five constraints across these four transitions because one transition—the statistical-to-neural shift—swaps two tightly coupled resources (labeled data, then the reusable representations learned from it); we treat that pair as one joint swap and say so explicitly rather than pretending the two were cleanly sequential. Table 1 places eighteen milestones on this single frame; read down its *Constraint* and *Unlock* columns and the relay race is visible at a glance.

The strongest objection is that the chain is too neat to be real. Distributed representations, attention, large-scale pretraining, and instruction tuning overlap heavily in time and reinforce one another, so assigning each era a single distinct scarce resource looks like narrative convenience rather than mechanism. The most damaging version notes that BERT-era representations and GPT-era compute arrived almost together, which undermines any clean “representations, then compute” ordering and suggests the swaps are an artifact of how we have chosen to slice a continuous process.

We answer by walking the chain and showing the swap at each joint.

3.1 Hand-authoring labor: ELIZA, AIML, and the knowledge-engineering ceiling

The first paradigm decided what to say by hand. ELIZA’s rules, PARRY’s affect variables, and SHRDLU’s microworld planner were all authored explicitly, and the binding constraint was the human labor required to write and maintain them. The three systems differ instructively in *where* they spent that labor: ELIZA spent it on surface reflection rules that needed no world model at all, which is exactly why a few hundred lines sufficed and why the gap between its competence and its perceived understanding was so wide; PARRY spent it on a small bank of affective state variables that made its persona more consistent but no more grounded; and SHRDLU spent it on a hand-built symbolic model of a blocks microworld, buying genuine reference and planning at the cost of a domain it could never leave. Read through the lens, all three are the same system under one constraint—authoring labor—allocated to three different illusions of depth.

The knowledge-engineering era industrialized that labor rather than escaping it. AIML let authors template thousands of patterns so that one author’s effort covered far more inputs, and Watson’s *Jeopardy!* system assembled an ensemble of curated retrieval-and-scoring pipelines tuned for a bounded question-answering task [Ferrucci et al., 2010]. Both were real engineering achievements, and both hit the same ceiling: coverage grew roughly linearly with the effort poured in, so capability plateaued at the edge of what had been hand-specified and degraded sharply at that edge. The characteristic failure, canned brittleness, is the signature of an authoring-bound system—fluent and even impressive inside the authored region, abruptly incoherent just outside it. Crucially, the escape did not come from authoring *better*; it came from changing what produced the answers at all, which is the first instance of the survey’s pattern that a binding constraint is relieved by a new resource (learned parameters) rather than by a cleverer application of the old one (more rules).

3.2 Labeled data and reusable representations: word2vec to seq2seq

Between the hand-authored era and the neural one sat a genuinely statistical period that the relay-race frame should not skip: through the 1990s and 2000s, n -gram language models, statistical machine translation, and partially-observable-Markov-decision-process dialogue managers replaced hand-written rules with parameters estimated from labeled in-domain corpora. The binding constraint in that period was *labeled data*: capability scaled with how much task-specific annotated dialogue or parallel text could be collected, and the commercial voice assistants that productized it—Apple’s Siri, Amazon’s Alexa, and Microsoft’s Cortana—felt brittle precisely because their understanding was bounded by the labeled intents their builders could afford to curate. The escape was to stop labeling and start learning reusable representations. Distributed representations such as GloVe [Pennington et al., 2014] replaced hand-coded features with vectors learned from unlabeled corpus statistics, and the binding constraint moved from labeled data to the availability of *reusable representations*. Sequence-to-sequence models with attention [Gehring et al., 2017] then let a single architecture map input utterances to output utterances without per-domain rules, collapsing the hand-built pipeline of intents, slots, and templates into one learned function. The shift is easy to under-read as merely “neural networks got better,” but the binding-constraint lens locates it precisely: what changed was not that the field acquired a new theory of conversation but that it stopped paying the per-domain labeling tax and began amortizing one representation across many tasks.

Fluency rose sharply; grounding did not, and the era’s failure mode shifted from canned brittleness to statistical incoherence—plausible-sounding output that drifts off topic or contradicts itself, because the model optimizes for likely continuations rather than for true ones. The same period

exposes the residue from two directions at once. When a system learned unfiltered from user input it could fail socially rather than merely factually, as Microsoft’s Tay demonstrated within hours of release; and when the same neural machinery was pointed at an explicit empathy objective it produced XiaoIce, a mass-consumer system whose design target was *perceived* rapport rather than factual grounding [Shum et al., 2018]. XiaoIce is the pivotal case for the survey’s thesis about the residue: it shows the field learning, deliberately and at scale, to manufacture the appearance of understanding as a product feature—industrializing the very gap ELIZA exposed rather than closing it. The statistical era thus relaxed the labeled-data and representation constraints while leaving grounding untouched, setting up the next handoff: representations were now cheap, so the question became what one could do with enough compute to train them on everything.

3.3 Compute and web text: the pretraining earthquake

Once representations were cheap, the binding constraint became raw compute and web-scale text. The Transformer mattered here for a reason that is structural rather than merely architectural: by making attention the whole computation it made pretraining embarrassingly parallel, converting the problem of learning language from one bounded by sequential recurrence into one bounded by how many accelerators a lab could point at how much text. That re-bounding is the whole event. The GPT line then scaled the recipe until few-shot competence emerged from next-token prediction alone [Floridi and Chiriatti, 2020]—a capability nobody hand-specified and that appeared only past a scale threshold, which is the signature of a substrate-gated unlock rather than an idea-gated one.

The same recipe transfers without modification to new corpora, and the consequences are diagnostic. Pointed at source code it produced systems useful enough to ship as developer tools, while exhibiting subtle, confident errors that a non-expert cannot easily catch [Chen et al., 2021]—the residue reappearing in a high-stakes domain, now wearing the costume of working software. The lesson the lens draws is that the pretraining earthquake bought breadth and fluency in direct proportion to compute and corpus, and bought grounding not at all: the same scaling that made GPT-3 fluent made its hallucinations fluent too. The unlock here is unambiguously a substrate—compute plus corpus—not a new theory of conversation, and the cleanest way to see this is counterfactual: the representations and the sequence-to-sequence machinery were already in hand for years, and capability did not jump until the compute to train large Transformers on web-scale text arrived. We sharpen exactly this argument, and defend it against the charge that it erases the Transformer’s genuine novelty, in §4.

3.4 Human-preference feedback: the alignment turn

A pretrained model is fluent but neither helpful nor safe by default, and the next scarce resource was *human-preference feedback* at scale. This transition is the sharpest test of the survey’s “substrate, not ideas” reading because the enabling *algorithm*—optimizing a learned reward model from human comparisons—was demonstrated years before it produced any mass-market assistant [Christiano et al., 2017]. The idea sat available and largely unused; what was missing was the industrial pipeline to collect, clean, and curate preference data at the scale a frontier model needs. When that pipeline was built, the same algorithm that had existed for years suddenly turned raw pretrained models into assistants—a textbook dormancy lag, and one we return to in §4.

The alignment era then generalized along three axes that are worth separating. Instruction finetuning showed that exposing a model to thousands of tasks phrased as instructions made the helpfulness effect generalize far beyond the tasks seen [Chung et al., 2022]; dialogue tuning, as in

Table 1: Cross-cutting comparison of eighteen consequential systems and milestones from ELIZA to the run-date frontier across the nine brief dimensions. The matrix is the table the whole survey points back to: reading down the *Binding constraint* and *Unlock* columns recovers the relay-race thesis. Cells that are non-comparable across eras are marked *n/c* rather than forced into a false equivalence. Citations anchor rows where a peer-reviewed corpus source exists; pre-2010 systems (ELIZA, PARRY, SHRDLU) and the run-date reasoning row are placed from primary system documentation and lab announcements and are flagged as such, not anchored to the closed corpus. Alignment-era rows cite the era’s instruction-tuning [Chung et al., 2022] and dialogue [Thoppilan et al., 2022] work in addition to the preference-RL roots [Christiano et al., 2017]; the “H800/H20-class” hardware entry on the reasoning row is a run-date observation, not a corpus-anchored figure.

System / Milestone	Paradigm (P)	Constraint (C)	Unlock (U)	Knowledge	Cap. vs perceived	Compute	Deployment	Failure
ELIZA (1966)	Pattern subst.	Authoring labor	Reflection rules	Hand-coded	Huge gap (ELIZA eff.)	Mainframe	Research demo	Canned brittleness
PARRY (1972)	Pattern + state	Authoring labor	Affect variables	Hand-coded	Large gap	Mainframe	Research demo	Rigid persona
SHRDLU (1972)	Symbolic plan	Authoring labor	Blocks microworld	Hand-coded KB	Gap (toy domain)	Mainframe	Research demo	No scaling out of world
A.L.I.C.E./AIML (2000s)	Knowledge eng.	Authoring labor	AIML templates	Curated rules	Lockstep-flattened	CPU	Web bot	Pattern ceiling
Watson (2011) [Ferrucci et al., 2010]	IR + ensemble	Labeled data/eng.	DeepQA pipeline	Curated corpora	Narrow, oversold	CPU cluster	Demo/enterprise	Brittle off-domain
word2vec/GloVe (2013-14) [Pennington et al., 2014]	Distrib. repr.	Representations	Embeddings	Corpus statistics	n/c (component)	GPU (small)	Library	No composition
seq2seq+attn (2014-17) [Gehring et al., 2017]	Neural transduction	Labeled pairs	Encoder-decoder	Training corpus	Fluent, shallow	GPU	Research/MT	Incoherence, drift
XiaoIce (2018) [Shum et al., 2018]	Neural + retrieval	In-domain data	Empathic ranking	Corpus + KB	Engineered rapport	GPU cluster	Mass consumer (CN)	Designed for perceived
BERT (2019)	Pretraining (enc.)	Compute + text	Bidir. pretrain	Web-scale corpus	Strong on NLU	GPU/TPU	Developer model	No generation
GPT-3 (2020) [Floridi and Chiariati, 2020]	Pretraining (dec.)	Compute + web text	Scale + few-shot	Web-scale	Fluent, ungrounded	Large GPU cluster	API	Hallucination
Codex (2021) [Chen et al., 2021]	Pretraining (code)	Compute + code data	Code corpus	Web + code	Useful, error-prone	GPU cluster	API/Copilot	Subtle code bugs
InstructGPT/RLHF [Christiano et al., 2017, Chung et al., 2022]	Pref. alignment	Human feedback	RLHF pipeline	Pretrain + prefs	Helpful, sycophantic	GPU cluster	API	Sycophancy
ChatGPT (2022) [Thoppilan et al., 2022]	Pref. alignment	Human feedback	Product + RLHF	Pretrain + prefs	Mass over-trust	Large cluster	Mass consumer	Confident wrongness
LaMDA (2022) [Thoppilan et al., 2022]	Pref. alignment	Human feedback	Dialog fine-tune	Pretrain + tools	“Sentience” panic	TPU cluster	Internal/demos	Anthropomorphism
Llama 2/3 (2023-24) [Touvron et al., 2023]	Open pretraining	Distribution	Open weights	Web-scale	Capable, open	GPU cluster	Open weights (US)	Safety variance
Mistral (2023) [Jiang et al., 2023]	Open + efficient	Distribution/eff.	GQA + SWA	Web-scale	Strong per-FLOP	GPU cluster	Open weights (EU)	Smaller-model limits
GLM/ERNIE (CN) [Du et al., 2022]	Pretraining (CN)	GPU access	Domestic stacks	Web-scale (CN)	Frontier-adjacent	Constrained GPUs	Closed/open (CN)	Export-control exposure
DeepSeek-RL-class (2025-26)	Reasoning/agnostic	Inference econ.	Test-time compute	Pretrain + tools	Strong, costly	H800/H20-class	API/open (CN)	Cost, confident error

LaMDA, made the assistant specifically conversational rather than merely responsive [Thoppilan et al., 2022]; and constitutional methods attacked the cost of the binding resource itself, seeking to reduce the human-labeling burden by sourcing alignment principles collectively or by having the model critique itself against a written standard rather than against one human label at a time [Huang et al., 2024]. That last move is telling: the era’s own research agenda was organized around *economizing on the scarce input*, which is precisely what the binding-constraint frame predicts a field will do once it recognizes what is binding. And the era’s characteristic failure, sycophancy and over-trust, is the residue re-emerging in its most pointed form: a system optimized for human approval is, by construction, optimized for the *perceived* side of the perceived-versus-actual gap, so the very technique that made models helpful also made their confident errors more persuasive.

We concede the boundaries blur, exactly as the steelman charges, and we owe the sharpest version—the near-simultaneity of representations and compute—an adjudication rather than a re-statement. Applying the counterfactual-leverage test of §2 at that joint: reusable representations [Pennington et al., 2014] were available and widely used for years before the pretraining earthquake, yet conversational capability did not jump until the compute to train large Transformers on web-scale text arrived—so at *this* joint the binding input at the margin was compute, not representations, because the representations were already present when capability was still blocked. That is a rebuttal, not a concession: the resources co-occurred in calendar time but not in leverage. Where even that test is genuinely indeterminate—some unlocks, such as instruction tuning, are part algorithm and part data-collection pipeline—we mark the relevant matrix cell *n/c* rather than forcing a single cause. The claim is about which input was binding at the margin in each era, not that history proceeded in clean, non-overlapping steps.

Laying the chain out end to end exposes a pattern no single paper could see: the unlocks were disproportionately substrate rather than conversational modeling ideas—which is the sharper, more contestable claim we turn to next.

4 Substrate, Not Ideas: The Dormancy-Lag Evidence

The decisive unlocks in conversational AI were disproportionately substrate—GPUs and CUDA, web-scale corpora, and preference-data pipelines—rather than conversational modeling ideas, and the evidence is a datable lag in which ideas that predated their enabling substrate lay dormant until the constraint they depended on was relaxed.

The strongest counterview is that this erases real algorithmic genius. The Transformer, chain-of-thought prompting, and learning from human preferences were genuine intellectual breakthroughs, and on this reading no quantity of compute would have produced a ChatGPT without them; ideas, not substrate, did the decisive work, and the hardware merely let good ideas run faster. A defender can add that substrate is endogenous—labs build clusters *because* they expect a known idea to pay off—so the causal arrow runs from ideas to substrate, reversing our claim.

We concede at the outset that ideas are necessary; the question is which input was *binding*, and the lags answer it.

4.1 The idea-to-payoff lag, measured

The pattern repeats across the chain. Neural sequence models were understood well before the compute to scale them existed; preference-based reinforcement learning was demonstrated around 2017 [Christiano et al., 2017] yet produced no mass-market aligned assistant until the preference-collection pipeline was built roughly five years later. Chain-of-thought is the case we must handle carefully, because it is easy to overclaim. Eliciting multi-step reasoning by prompting [Kojima et al., 2022], and improving it by sampling several chains and voting [Wang et al., 2022], was a genuine *conceptual* contribution—we concede that the prompting insight is an idea, not a substrate, and that calling it “not an idea because it needs no new architecture” would be the conflation our critics rightly warn against. We therefore narrow the substrate claim for this case: the conceptual insight was real, but what gated its *payoff at scale* was the inference compute to sample and vote over many chains, and that economics is substrate. The lag evidence concerns when a known capability cashed out, not who deserves intellectual credit for it. An idea that waits years for its substrate was not the binding constraint; an idea that needs more inference compute to pay off is gated by that compute.

The lag is not a single anecdote but a repeated shape, and stating it as a shape is what protects the claim from cherry-picking. In each instance the same three conditions hold: the idea was published and understood; the other inputs it needed were already present; and the conversational payoff nonetheless waited until one specific resource—compute, corpus, or a data-collection pipeline—arrived. Neural sequence modeling waited on the compute to train at scale; preference-based reinforcement learning [Christiano et al., 2017] waited on the preference-collection pipeline; few-shot competence [Floridi and Chiriatti, 2020] waited on web-scale pretraining compute. The frame makes a falsifiable demand of itself here: if any of these capabilities had jumped at the moment the *idea* appeared, with its enabling resource still absent, the substrate reading would be wrong for that case. It is the consistency of the lag—idea first, resource later, payoff with the resource—across otherwise unrelated transitions that turns a set of stories into evidence.

4.2 AlexNet/CUDA: the substrate origin story

The modern trajectory’s true origin is a hardware repurposing, not a conversational result. Gradient-trained deep networks were established in the 1990s [LeCun et al., 1998], but the field stalled until

general-purpose GPU compute made large networks trainable at useful scale—an exogenous shift driven by graphics and gaming economics, not by any conversational-AI agenda. Every subsequent “modeling” win, including the Transformer, was affordable only on the substrate that this repurposing created. Systems-level substrate advances continue to do load-bearing work that is invisible in the modeling literature: memory management techniques that make serving large models economical [Kwon et al., 2023] expand what can actually be deployed without changing the model at all.

This last point deserves emphasis because it is where the substrate argument is most often underestimated. The modeling literature naturally foregrounds architectures and training objectives, and treats serving as an implementation detail; but at the frontier the question of which conversational system hundreds of millions of people actually experience is decided as much by inference economics as by model design. A capability that exists only at a per-query cost no one can afford is, for the purposes of the history we are writing, a capability that has not yet been unlocked—and a serving advance that halves that cost [Kwon et al., 2023] can do more to change what conversational AI *is* in practice than a modeling tweak that nudges a benchmark. The substrate, in other words, is not only what makes a capability trainable; it is also what makes a trained capability deployable, and both edges have repeatedly been the binding ones.

4.3 Did ideas pull the substrate? Adjudicating reverse causality

We grant partial endogeneity: some clusters were indeed built in anticipation of payoff. But the reverse-causality story breaks at the exogenous shocks. The GPU’s suitability for deep learning, the gaming market that made GPUs cheap, and later the state export controls of §5 are substrate changes *not* driven by conversational-AI ideas, and capability tracks them. When the substrate moved for reasons external to the field, the field moved with it; that is the signature of a binding input.

The endogeneity objection, properly stated, is that clusters are built in anticipation of payoff, so substrate is downstream of belief in an idea and the causal arrow runs from ideas to hardware. We grant this for the *within*-field case and still find the arrow points our way at the margins that matter, for a structural reason: the largest substrate shifts in the history were not commissioned by conversational-AI labs at all. The suitability of the GPU for dense linear algebra was a by-product of graphics and gaming economics; the collapse in the price of that compute was driven by a consumer market with no interest in dialogue; and the export-control regime of §5 is a state-security intervention, not a research decision. These are exogenous shocks to the field’s input supply, and conversational capability tracks them. A reverse-causal story in which ideas pull substrate cannot explain why capability moves in lockstep with hardware changes that no one in the field asked for—which is exactly the discriminating evidence the endogeneity worry would predict we could not find.

We concede this is an argument about emphasis, not a denial, and the honest test of that is to name a transition where the binding input was plausibly an *idea*, not substrate—and let it stand. The Transformer’s self-attention is that case. Attention reorganized computation in a way that no quantity of additional hardware applied to recurrent architectures would have produced on its own; the idea was load-bearing, and on existing GPUs it paid off almost immediately rather than lying dormant. We do not explain this away. What we claim is narrower than “ideas never bind”: it is that *across the sweep*, the resource whose absence most often blocked capability was substrate rather than idea, and that the dormancy lags are the evidence for the typical case. The

self-attention counterexample shows the thesis is about the modal transition, not a law without exceptions—which is why we state it as relocation of the *binding* constraint, not as the irrelevance of ideas. We separate two joints the critic may otherwise conflate: self-attention’s payoff at 2017-era scale was idea-bound and immediate, but the assistant-grade conversational capability it eventually enabled—few-shot competence emerging from next-token prediction [Floridi and Chiriatti, 2020]—came years later and waited on web-scale compute, exactly as §3 argues. The Transformer is a load-bearing idea; the conversational capability it ultimately underwrote was still substrate-gated, so conceding the first does not concede the second.

If substrate is the binding input, then capability should track hardware and data step-changes more tightly than algorithmic milestones—a prediction that becomes decisive at the run-date frontier, where the substrate is contested by states.

5 Compute and Geopolitics: The Constraint Leaves the Model

At the run-date frontier the binding constraint has left the model architecture altogether: the two scarce resources now throttling who can build a frontier conversational system are training-data exhaustion and access to compute—export-controlled accelerators on the training side and inference economics on the serving side, which we treat as two faces of the single compute-access constraint—and the clearest evidence is that the US–China frontier race has formed precisely along the GPU-access fault line rather than along any line of modeling talent.

The strongest objection is that the Chinese frontier disproves the claim. Labs operating under US export controls have shipped competitive open-weight models and frontier-class reasoning systems, apparently on the strength of efficiency and algorithmic ingenuity rather than raw compute. If a compute-constrained actor can match the frontier, then ideas, not substrate, are doing the decisive work, and the entire geopolitics layer is a sideshow to continued algorithmic progress.

We engage this directly rather than dodging it, and the evidence still favors the constraint-relocation reading.

5.1 The export-control natural experiment

The 2022 and 2023 US controls on advanced accelerators are the cleanest natural experiment available for what governs conversational capability. If ideas were the binding input, restricting hardware would be futile, since ideas diffuse across borders within days of publication. That two governments instead chose to contest the frontier by restricting *compute*—and that vendors responded with deliberately throttled parts (the A800, then H800, then H20 lineage) engineered to sit just under the control thresholds—is a revealed-preference signal that the actors with the most at stake believe the substrate, not the idea, is the lever. Analyses of semiconductor industrial policy frame exactly this dynamic [Bown and Wang, 2024].

What makes this a natural experiment rather than a mere illustration is that the intervention acts on exactly one of the binding-constraint candidates while leaving the other untouched. Ideas, papers, and trained-model techniques continue to diffuse across the border essentially without friction; what the controls restrict is the physical substrate. If conversational capability were idea-bound, an intervention that throttles only hardware should have little effect on who can reach the frontier, and the rational response of a constrained actor would be to lean harder on its undiminished access to ideas. Instead the constrained actors pour effort into efficiency, work-around parts, and domestic-silicon programs—that is, into manufacturing access to the restricted resource—which is the behavior the binding-constraint frame predicts and the idea-bound reading does not. The

controls, in other words, are a costly bet by the actors with the most information and the most at stake that compute is the lever, and their adversaries’ response is a second, independent vote for the same proposition.

5.2 The NVIDIA line as a capability proxy

Across the pretraining and alignment eras, frontier conversational capability appears to have tracked the datacenter accelerator line—V100, A100, H100/H200, and Blackwell—at least as tightly as it tracked any single algorithmic milestone, because each generation changed what scale of training and what economics of serving were affordable. We state this as a structural observation continuous with the substrate argument of §4 rather than as a measured correlation: we have not regressed a capability index on accelerator generation, and a reader should read the claim as the hypothesis whose forward test is FD3’s kill-criterion, not as an established quantitative result. Memory and serving advances that let a fixed model be deployed more cheaply [Kwon et al., 2023] belong to the same substrate story: at the frontier, inference economics decides what can be offered to hundreds of millions of users, independent of the model’s design.

5.3 The run-date frontier: confirming SOTA, superseding the stale

The brief that commissioned this survey demanded the frontier be established *empirically* at run time rather than frozen at an earlier snapshot, and several version numbers that were current when the brief was written are already stale. We are therefore explicit about evidentiary status: the specific model names, version numbers, and leaderboard positions in this subsection are *run-date observations*—drawn from lab announcements, model cards, and the public human-preference arena as read at the run date—and are *not* anchored to the peer-reviewed closed-set corpus, which by construction lags the frontier by months. We flag them as such rather than dressing them in decorative citations to older papers. As of the run date, the flagship line is observed to include OpenAI’s GPT-5.5 line, Anthropic’s Claude Opus 4.8, and Google’s Gemini 3.1 Pro, with xAI’s Grok and Meta’s Llama 4 line in the same tier and an open-weight wave from Mistral. The Chinese frontier—DeepSeek’s V3/V4-class successors, Alibaba’s Qwen3, Zhipu’s GLM line, Moonshot’s Kimi, MiniMax, and the latest Ernie, Doubao, and Hunyuan—is observed to sit close behind the US flagships despite operating on export-constrained accelerators, competing hardest on cost and on Chinese-language tasks. Where the closed corpus does support a structural claim we cite it. The open Llama line is documented in Touvron et al. [2023] and Mistral’s efficient open-weight design in Jiang et al. [2023]. On the Chinese side, the GLM line is documented in Du et al. [2022] and the earlier ERNIE representation work in Zhang et al. [2019], while broad multimodality is now a standard capability [Team et al., 2023]. The structural point survives the version churn: a US lab ties a flagship release to a specific hardware generation, while Chinese labs route around chip restrictions, and the competition is organized around compute access. We also note, as a conjecture rather than a finding, that lines which have gone quiet appear more often to have lost compute access or funding than to have run out of ideas; we cannot establish this from the closed corpus and flag it as an untested observation for the trajectory analysis.

5.4 Data exhaustion: the second terminal constraint

The thesis names *two* resources that have escaped the model—compute access and data exhaustion—and we treat the second with the same discipline as the first rather than as an afterthought. The

binding-leverage test applies cleanly: high-quality human text is a non-renewable input whose depletion blocks the next pretraining-scale jump *even when compute is abundant*, which is precisely the counterfactual signature of a binding constraint. The evidence that it now binds is that frontier labs have turned to synthetic generation, self-improvement, and aggressive filtering—behaviors that only make sense once the cheap, high-quality human corpus is substantially consumed—and that the open risk of these substitutes is model collapse when systems train on their own outputs, which we develop as Problem 2 (§7) and forecast under FD2. Inference economics is not a third independent constraint but the serving-side face of compute access: reasoning-style systems explicitly trade more inference compute for accuracy—sampling and voting over many chains is precisely spending inference compute to buy reliability [Wang et al., 2022]—and the cost of serving a good answer to hundreds of millions of users is itself a function of accelerator supply [Kwon et al., 2023]. So the operative limit at the frontier is the *cost* of a good answer, not the existence of a good model, and that cost folds back into the compute-access constraint. Data exhaustion and compute access are thus the two terminal constraints the thesis names—the inside and the outside of the same post-architecture bottleneck—and FD2’s kill-criterion gives the data half the same falsifiable weight as FD3 gives the compute half.

We concede that efficiency results show genuine partial substitution: a constrained actor can convert ingenuity into capability and narrow the gap, and a strong-form “compute is destiny” claim is unsupported. We therefore mark the substitution coefficient—how far efficiency can offset raw compute—as a real open question and treat the next twelve to thirty-six months as the live test. But the constrained labs still depend on accumulated and work-around accelerators, and their efficiency race is itself a *response* to a binding compute constraint, which is exactly what the substrate frame predicts a constrained actor will do.

If the binding constraint has relocated to compute access, then the run-date SOTA distribution should track accelerator supply and capital expenditure more tightly than research output—the survey’s central forward claim, whose kill-criterion we state explicitly in the closing sections.

6 Evaluation: How Each Era Judged “Good Conversation”—and Why Each Measure Broke

Each era’s evaluation regime measured a proxy for that era’s binding constraint and saturated or was gamed the moment the constraint was relaxed—from the Turing Test and Loebner Prize, through perplexity and BLEU, through static benchmarks such as GLUE, SQuAD, and MMLU, to human-preference arenas—so no measure is ground truth, the residue (the perceived-versus-actual gap) survives every regime, and this recurring saturation is itself evidence that the binding-constraint frame out-predicts the naive “models got smarter” narrative.

A defender of benchmarks argues that measurement genuinely improved over time. The progression from GLUE to SuperGLUE to MMLU to live arenas is, on this view, a story of increasingly robust and better-correlated evaluation, and saturation simply means the field solved the task it set—hardly an indictment of the metric. The strongest form notes that human-preference arenas capture something real that static benchmarks miss, so evaluation is converging on truth, not chasing its tail.

The history shows each regime measuring perceived capability and breaking when its era ended.

6.1 Turing Test and Loebner: measuring perceived humanness

The founding measure asked whether a machine could be *mistaken* for a human—a test of perceived humanness, not of capability or understanding. This was not a flaw in the test so much as an honest reflection of what the era could measure: with no large labeled corpora and no agreed task suites, the only available yardstick was a human judge’s impression, and an impression is precisely a measurement of the perceived side of the residue. The Loebner Prize operationalized the test and, predictably, was promptly gamed by systems optimized for deception rather than competence—misdirection, canned humor, deliberate typos—which is why “passing the Turing Test” became a folklore claim to be verified rather than a meaningful capability milestone.

The deeper point for the survey is that the founding measure scored the residue *directly*, and did so honestly, because in the hand-authoring era perceived understanding was almost the only thing a system had. ELIZA’s effect and the Loebner Prize’s gameability are the same phenomenon viewed from the system side and the evaluator side: a metric that rewards the appearance of understanding will be won by whatever most cheaply produces that appearance. Every later regime inherits this hazard the moment it, too, starts scoring a proxy for perceived quality—a continuity we now trace through the static-benchmark and arena eras.

6.2 Perplexity/BLEU to GLUE/SQuAD/MMLU: the static-benchmark era and its saturation

As the field became data- and representation-bound, evaluation became corpus-based. Perplexity and BLEU measured fluency against references; then task-suite benchmarks measured language understanding. GLUE was introduced as a multi-task understanding benchmark [Wang et al., 2018], was solved fast enough to require the deliberately harder SuperGLUE [Wang et al., 2019], and the reading-comprehension line responded to models exploiting answerable-question artifacts by adding unanswerable questions in SQuAD 2.0 [Rajpurkar et al., 2018]. The pattern is uniform: each static benchmark saturated, then leaked into training data, the moment the binding constraint it indexed was relaxed. The GLUE-to-SuperGLUE-to-SQuAD 2.0 sequence is the clearest illustration of the arms race the lens predicts: a benchmark indexes the capability that is currently scarce, models acquire that capability as the era’s constraint is removed, the benchmark saturates, and a deliberately harder successor is built to re-open the gap—only to saturate in turn. SQuAD 2.0’s specific innovation, adding unanswerable questions so that a model must decline rather than always produce an answer [Rajpurkar et al., 2018], is itself a direct attempt to measure a sliver of the residue: it scores whether a system knows when it does *not* know, which is exactly the robustness dimension that fluency hides.

Holistic multi-metric evaluation [Bommasani et al., 2023] was the era’s most serious attempt to outrun saturation, by measuring many capabilities and many harms at once rather than a single score. But breadth and contamination are orthogonal problems: widening the battery of tasks raises the cost of gaming any one of them, yet does nothing about the fact that once a benchmark is public it leaks into the next model’s training corpus, after which a high score no longer distinguishes understanding from memorization. The static-benchmark era thus ends where it began—measuring a proxy that decays the moment its era’s constraint is relaxed—which is what pushes evaluation toward the live, preference-based arenas of the alignment era.

6.3 Human-preference arenas: Elo, MT-Bench, and what they game

When models became preference-aligned, evaluation followed the constraint to human preference itself: pairwise arena comparisons and model-graded benchmarks such as MT-Bench scored which answer people (or a strong model proxy) preferred [Zheng et al., 2023]. These capture helpfulness that static suites miss, but they measure *preference*, which is gameable by formatting, verbosity, and sycophancy—the very behaviors the alignment era’s failure mode names. The coincidence is not accidental: the arena measures the same quantity the alignment era optimizes, so the evaluation regime and the training objective share a blind spot. A model tuned to be preferred and an arena that scores preference will agree with each other about quality precisely where both are wrong about robustness, which means the newest and most realistic-seeming instrument reproduces the ELIZA effect at the level of the measurement itself.

The arena therefore marks the end state of a long regression rather than an arrival at ground truth. The founding measure scored perceived humanness; the static era scored proxies that leaked; the arena scores perceived helpfulness. At no point did the field acquire an instrument that scores the residue directly—the signed gap between what a system can reliably do and what its users believe it can do—and the model-graded variant [Zheng et al., 2023], while scalable, simply moves the perceived-quality judgment from a human to another model trained on human preferences, inheriting the same bias one level up. This is why we argue in §7 that the next durable measure must be built to track the residue on purpose, because every measure built to track an era’s binding-constraint proxy has saturated on schedule.

We concede that each regime was a real improvement on its predecessor and that arenas genuinely capture something static benchmarks cannot. The point is not that evaluation is worthless but that every regime measured a proxy for perceived capability and broke under saturation or contamination, never closing the residue; better measurement has not yet become measurement of understanding.

That every evaluation regime saturated in lockstep with the removal of its era’s binding constraint is the predictive payoff of the frame: it tells us the next durable measure must track the residue rather than the current proxy, which is the first of the open problems we now state.

7 Open Problems

7.1 The perceived-versus-actual understanding gap persists

The residue ELIZA exposed in 1966—users attributing understanding in proportion to fluency rather than to demonstrated robustness—has survived every paradigm and is, at the frontier, essentially unmeasured. Retrieval, tool use, and preference alignment reduce some error classes and raise apparent reliability, and arena evaluation [Zheng et al., 2023] captures user satisfaction. None of these measures the gap itself; preference-tuned systems can widen it by producing confident, well-formatted wrongness that non-experts cannot detect. We need contamination-controlled instruments that score the gap directly. Concretely, define *attributed capability* as the success rate users or expert judges *predict* on held-out items before seeing outcomes, and *actual capability* as measured accuracy on the same items under contamination control; the residue is the signed gap between them. Preference arenas [Zheng et al., 2023] measure the wrong side of this gap—they score perceived quality—so a residue-tracking instrument must pair them with a held-out reliability probe in the spirit of unanswerable-question evaluation [Rajpurkar et al., 2018]. This makes the problem measurable rather than aspirational (paired with FD1).

7.2 Training-data exhaustion

High-quality human text is a depleting resource, and the pretraining era’s core substrate is running out. Synthetic data, self-improvement, and aggressive filtering extend the corpus. Synthetic-data substitution is unproven at frontier scale and risks model collapse when models train on their own outputs. Data-efficiency regimes and non-degenerate synthetic pipelines are the open frontier. The first tractable target is a measured *collapse threshold*: the synthetic-to-human mixing ratio at which held-out, contamination-controlled accuracy begins to degrade across successive training generations—turning “provably non-degenerate” from a slogan into a curve a lab can chart (paired with FD2).

7.3 The compute-access split

Export controls and accelerator scarcity may bifurcate the frontier along geopolitical lines, decoupling capability from research merit [Bown and Wang, 2024]. Efficiency innovation, work-around accelerators, and domestic-silicon programs partially substitute for restricted compute. The degree of substitution is unknown, and a persistent split would make “who can build a frontier assistant” a question of hardware policy, not science. Measuring the compute-to-capability substitution coefficient over time is the decisive empirical question (paired with FD3).

7.4 Evaluation that survives saturation

Every measurement regime to date has been gamed or saturated once its era’s constraint was removed (§6). Holistic suites [Bommasani et al., 2023], harder successors [Wang et al., 2019], and live arenas each attempt to outrun saturation. Breadth does not solve contamination, and preference arenas measure a gameable proxy. Adversarial, live, contamination-proof evaluation tied to the residue rather than to the current proxy is required (paired with FD4).

7.5 Whether reasoning is a new paradigm or a compute trade

It is unresolved whether test-time-compute “reasoning” is a genuine new paradigm or another substrate move—spending more inference compute for accuracy. Chain-of-thought [Kojima et al., 2022] and self-consistency [Wang et al., 2022] elicit reasoning from existing models without new architectures. That these require no architectural change is precisely what makes the paradigm-versus-substrate question hard to settle. Isolating whether reasoning gains scale with inference spend or with a new learning principle would adjudicate it (paired with FD5).

7.6 Grounding and hallucination under deployment

Factual grounding at scale is unsolved; deployed systems produce fluent, confident falsehoods. Retrieval augmentation and tool use reduce some hallucinations; serving advances make grounding affordable [Kwon et al., 2023]. Grounding layers reduce but do not eliminate confident wrongness, and coverage of the long tail remains poor. Verifiable-provenance generation, where every assertion carries a checkable source, is the promising direction. Operationalize it as a *provenance-coverage* metric—the fraction of model assertions that carry a checkable source—scored on a held-out reliability probe in the unanswerable-question spirit [Rajpurkar et al., 2018], with serving cost [Kwon et al., 2023] reported alongside so that provenance is not bought at intractable latency (paired with FD6).

8 Future Directions

FD1 — Robustness–attribution alignment. We predict that, through 2028, contamination-controlled out-of-distribution robustness will continue to lag the capability users attribute to frontier systems—the ELIZA residue will remain open under honest measurement. We make this measurable by the operationalization of Problem 1: attributed capability as held-out, pre-outcome success predictions by users or expert judges, and actual capability as measured contamination-controlled accuracy on the same items. **Kill-criterion:** a frontier model whose measured OOD reliability matches its attributed capability under this paired, contamination-controlled protocol falsifies this.

FD2 — Synthetic-data and data-efficiency regimes. We predict the next capability jump comes from data-efficiency and synthetic-data pipelines, not from a new architecture, because the binding constraint is text exhaustion rather than model design. **Kill-criterion:** a frontier leap driven by a novel architecture trained on the same or smaller data budget than its predecessor falsifies this.

FD3 — Compute-access as the frontier’s organizing axis. We predict that run-date state of the art will continue to track accelerator supply and capital expenditure more tightly than research output through 2027, with the US–China gap moving as compute access moves. To keep this falsifiable without making it vacuous, we define *compute-poor* as a disjunction at the time of the leap: a lab is compute-poor if it has *either* no lawful access to current-generation frontier accelerators *or* a total training-plus-serving fleet whose FLOP-normalized capacity (counting actual deployed hardware, converted to equivalents of a named current-generation part) is below the scale of contemporaneous frontier training runs. The two-pronged form matters: a pure absolute-count bar would be self-defeating, since leading the frontier currently *requires* tens of thousands of accelerator-equivalents, so the access prong is what lets a constrained-but-ingenious win actually trigger the criterion. By this definition the run-date Chinese frontier is *constrained but not compute-poor*: labs such as those behind the GLM line [Du et al., 2022] are excluded on the access prong (they operate large prior-generation and work-around H800/H20 fleets), not by an elastic recount of equivalents. **Kill-criterion:** a lab meeting either prong of the compute-poor definition reproducibly leading the frontier on the strength of a modeling idea alone—also the kill-criterion for the survey’s central thesis.

FD4 — Adversarial, live, contamination-proof evaluation. We predict the field abandons static benchmarks as its headline measure in favor of live, adversarial, preference-and-task evaluation, because every static suite has saturated. **Kill-criterion:** a static benchmark that resists saturation and contamination for more than three years would falsify the need for the shift.

FD5 — Reasoning as substrate, not paradigm. We predict that “reasoning” gains will scale with inference-compute spend rather than with a new learning theory, marking the reasoning turn as another substrate move. **Kill-criterion:** a reasoning advance that delivers frontier gains at flat or reduced inference cost would falsify this.

FD6 — Grounded generation with verifiable provenance. We predict that factuality gains come from provenance and verification layers rather than from scale alone, because scale has not

closed the hallucination gap. **Kill-criterion:** hallucination driven to negligible rates by model scale alone, with no grounding or verification layer, would falsify this.

9 Conclusion: The Verdict

Seventy-six years of conversational machines look, from a distance, like one long climb toward intelligence. Read up close, through the binding-constraint lens, they are something stranger and more useful to understand: a relay race in which each era won not by becoming globally smarter but by removing the one resource its predecessor was starved of—hand-authoring labor, then labeled data, then reusable representations, then raw compute and web text, then human-preference feedback—and in which the baton has now passed out of the model entirely, into data exhaustion and the geopolitics of compute. That is the survey’s verdict, and we have tried to earn it rather than assert it.

The chain is real, and it is substrate-shaped. Laid end to end (§3, Table 1), the transitions resolve into constraint swaps whose unlocks were more often substrate—GPUs, corpora, preference pipelines—than conversational modeling ideas. The dormancy lags (§4) are the decisive evidence: ideas that waited years for their enabling resource, from neural language models to learning from human preferences to chain-of-thought, were not the binding constraint, however brilliant.

The frontier is drawn by hardware, not by ideas. The export-control natural experiment (§5) shows states contesting conversation by restricting compute, because ideas diffuse and accelerators do not. We concede what the steelman demands—efficiency can partially substitute for raw compute, and the substitution coefficient is genuinely unknown—but the constrained labs’ very strategy is a response to a binding compute constraint, which is what the frame predicts.

The residue was never closed; it was industrialized. The gap between perceived and actual understanding that ELIZA exposed reappears, at scale, in every era’s failure mode and inside every era’s evaluation regime (§6). The field did not solve understanding; it learned to mass-produce its appearance, and each measurement instrument saturated as soon as its era’s constraint was relaxed.

Limitations, honestly stated. Three are worth naming. Our scope is text-first, so the voice-assistant wave (Siri, Alexa, Cortana) and incidents like Tay appear as evidence rather than as objects of study in their own right. Our cross-cutting matrix is hand-curated and carries survivorship risk: it records the systems that mattered enough to remember, which can make the trajectory look tidier than it was. And a critic may charge that our central kill-criterion is gerrymandered—that by classing the export-constrained Chinese frontier as “constrained but not compute-poor” we define the one live disconfirming case out of existence. We have tried to defuse this honestly: the compute-poor definition (§8) is a disjunction with an *access* prong precisely so that an ingenious win under genuine hardware denial *can* fire it. If the reader still finds the carve-out too convenient, that is the most important place to test the thesis.

A call to action, and the conditions for being wrong. If the frame is right, the field should stop measuring the current proxy and start measuring the residue, should treat data and inference economics as first-class research problems rather than engineering footnotes, and should expect the next leaders to be set by compute access as much as by insight. We have attached a kill-criterion to every forward claim (§8); the load-bearing one is now operational. If, within the next thirty-six months, a *compute-poor* lab—one that either has no lawful access to current-generation frontier accelerators or fields a total fleet below contemporaneous frontier-training scale, a bar the export-constrained-but-well-provisioned Chinese frontier does not meet on the access prong—reproducibly leaps the frontier on the strength of a modeling idea alone, then the binding constraint never left

the model, and this survey is wrong. We think it will not happen—but we have said, in advance, exactly what would change our minds.

References

- R. Bommasani, P. Liang, and T. Lee. Holistic evaluation of language models. *Annals of the New York Academy of Sciences*, 2023. doi: 10.1111/nyas.15007. URL <https://onlinelibrary.wiley.com/doi/pdfdirect/10.1111/nyas.15007>.
- C. P. Bown and D. Wang. Semiconductors and modern industrial policy. *The Journal of Economic Perspectives*, 2024. doi: 10.1257/jep.38.4.81. URL <https://www.aeaweb.org/articles/pdf/doi/10.1257/jep.38.4.81>.
- Y. Chang, X. Wang, J. Wang, Y. Wu, L. Yang, K. Zhu, H. Chen, X. Yi, C. Wang, Y. Wang, W. Ye, Y. Zhang, Y. Chang, P. S. Yu, Q. Yang, and X. Xie. A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology*, 2024. doi: 10.1145/3641289. URL <https://doi.org/10.1145/3641289>.
- M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. de Oliveira Pinto, J. Kaplan, H. Edwards, Y. Burda, N. Joseph, G. Brockman, A. Ray, R. Puri, G. Krueger, M. Petrov, H. Khlaaf, G. Sastry, P. Mishkin, B. Chan, S. Gray, N. Ryder, M. Pavlov, A. Power, Ł. Kaiser, M. Bavarian, C. Winter, P. Tillet, F. P. Such, D. Cummings, M. Plappert, F. Chantzis, E. A. Barnes, A. Herbert-Voss, W. H. Guss, A. Nichol, A. Paino, N. Tezak, J. Tang, I. Babuschkin, S. Balaji, S. Jain, W. S. Saunders, C. Hesse, A. N. Carr, J. Leike, J. Achiam, V. Misra, E. Morikawa, A. Radford, M. M. Knight, M. Brundage, M. Murati, K. Mayer, P. Welinder, B. McGrew, D. Amodei, S. McCandlish, I. Sutskever, and W. Zaremba. Evaluating large language models trained on code. *arXiv (Cornell University)*, 2021. doi: 10.48550/arxiv.2107.03374. URL <https://arxiv.org/pdf/2107.03374>.
- P. F. Christiano, J. Leike, T. B. Brown, M. Martic, S. Legg, and D. Amodei. Deep reinforcement learning from human preferences. *arXiv (Cornell University)*, 2017. doi: 10.48550/arxiv.1706.03741. URL <https://arxiv.org/pdf/1706.03741>.
- H. W. Chung, L. Hou, S. Longpre, B. Zoph, Y. Tay, W. Fedus, E. Li, X. Wang, M. Dehghani, S. Brahma, A. Webson, S. Gu, Z. Dai, M. Süzgün, X. Chen, A. Chowdhery, A. Castro-Ros, M. Pellat, K. Robinson, D. Valter, S. Narang, G. Mishra, A. Yu, V. Zhao, Y. Huang, A. M. Dai, H. Yu, S. Petrov, E. H., J. Dean, J. Devlin, A. Roberts, D. Zhou, Q. V. Le, and J. Wei. Scaling instruction-finetuned language models. *arXiv (Cornell University)*, 2022. doi: 10.48550/arxiv.2210.11416. URL <https://arxiv.org/pdf/2210.11416>.
- Z. Du, Y. Qian, X. Liu, M. Ding, J. Qiu, Z. Yang, and J. Tang. Glm: General language model pretraining with autoregressive blank infilling. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022. doi: 10.18653/v1/2022.acl-long.26. URL <https://aclanthology.org/2022.acl-long.26.pdf>.
- D. Ferrucci, E. W. Brown, J. ChuCarroll, J. Fan, D. Gondek, A. Kalyanpur, A. Lally, J. Murdock, E. Nyberg, J. Prager, N. Schlaefer, and C. Welty. Building watson: An overview of the deepqa project. *AI Magazine*, 2010. doi: 10.1609/aimag.v31i3.2303. URL <https://ojs.aaai.org/index.php/aimagazine/article/download/2303/2165>.

- L. Floridi and M. Chiriatti. Gpt-3: Its nature, scope, limits, and consequences. *Minds and Machines*, 2020. doi: 10.1007/s11023-020-09548-1. URL <https://link.springer.com/content/pdf/10.1007/s11023-020-09548-1.pdf>.
- J. Gehring, M. Auli, D. Grangier, D. Yarats, and Y. Dauphin. Convolutional sequence to sequence learning. *arXiv (Cornell University)*, 2017. doi: 10.48550/arxiv.1705.03122. URL <https://arxiv.org/pdf/1705.03122>.
- S. Huang, D. Siddarth, L. Lovitt, T. I. Liao, E. Durmus, A. Tamkin, and D. Ganguli. Collective constitutional ai: Aligning a language model with public input. *Unknown*, 2024. doi: 10.1145/3630106.3658979. URL <https://dl.acm.org/doi/pdf/10.1145/3630106.3658979>.
- A. Q. Jiang, A. Sablayrolles, A. Mensch, C. Bamford, D. S. Chaplot, D. de Las Casas, F. Bressand, G. Lengyel, G. Lample, L. Saulnier, L. R. Lavaud, M.-A. Lachaux, P. Stock, T. L. Scao, T. Lavril, T. J. Wang, T. Lacroix, and W. E. Sayed. Mistral 7b. *arXiv (Cornell University)*, 2023. doi: 10.48550/arxiv.2310.06825. URL <https://arxiv.org/pdf/2310.06825>.
- T. Kojima, S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa. Large language models are zero-shot reasoners. *arXiv (Cornell University)*, 2022. doi: 10.48550/arxiv.2205.11916. URL <https://arxiv.org/pdf/2205.11916>.
- W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, and I. Stoica. Efficient memory management for large language model serving with pagedattention. *Unknown*, 2023. doi: 10.1145/3600006.3613165. URL <https://dl.acm.org/doi/pdf/10.1145/3600006.3613165>.
- Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner. Gradient-based learning applied to document recognition. In *Proceedings of the IEEE*, 1998. doi: 10.1109/5.726791. URL <https://hal.science/hal-03926082/document>.
- J. Moulal. The chinese room argument. *Cambridge University Press eBooks*, 2003. doi: 10.1017/cbo9780511613999.010. URL <https://doi.org/10.1017/cbo9780511613999.010>.
- J. Pennington, R. Socher, and C. D. Manning. Glove: Global vectors for word representation. *Unknown*, 2014. doi: 10.3115/v1/d14-1162. URL <https://doi.org/10.3115/v1/d14-1162>.
- M. A. K. Raiaan, M. S. H. Mukta, K. Fatema, N. M. Fahad, S. Sakib, M. M. J. Mim, J. Ahmad, M. E. Ali, and S. Azam. A review on large language models: Architectures, applications, taxonomies, open issues and challenges. *IEEE Access*, 2024. doi: 10.1109/access.2024.3365742. URL <https://ieeexplore.ieee.org/ielx7/6287639/6514899/10433480.pdf>.
- P. Rajpurkar, R. Jia, and P. Liang. Know what you dont know: Unanswerable questions for squad. *Unknown*, 2018. doi: 10.18653/v1/p18-2124. URL <https://www.aclweb.org/anthology/P18-2124.pdf>.
- K. I. Roumeliotis and N. D. Tselikas. Chatgpt and open-ai models: A preliminary review. *Future Internet*, 2023. doi: 10.3390/fi15060192. URL <https://www.mdpi.com/1999-5903/15/6/192/pdf?version=1685089354>.

- H. Shum, X. He, and D. Li. From eliza to xiaoice: challenges and opportunities with social chatbots. *Frontiers of Information Technology & Electronic Engineering*, 2018. doi: 10.1631/fitee.1700826. URL <https://doi.org/10.1631/fitee.1700826>.
- G. R. Team, R. Anil, S. Borgeaud, J.-B. Alayrac, J. Yu, R. Soricut, J. Schalkwyk, A. M. Dai, A. Hauth, K. Millican, D. M. Silver, M. Johnson, I. Antonoglou, J. Schrittwieser, A. Glaese, J. Chen, E. Pitler, T. Lillicrap, A. Lazaridou, O. Firat, J. Molloy, M. Isard, P. R. Barham, T. Hennigan, B. Lee, F. Viola, M. Reynolds, Y. Xu, R. C. Doherty, E. Collins, C. Meyer, E. Rutherford, Érica Rodrigues Moreira, K. Ayoub, M. Goel, J. Krawczyk, C. Du, E. H., H.-T. Cheng, E. C. Ni, P. Shah, P. Kane, B. Chan, M. Faruqui, A. Severyn, H. Lin, Y. Li, Y. Cheng, A. Ittycheriah, M. Mahdieh, M. Chen, P. Sun, D. Tran, S. Bagri, B. Lakshminarayanan, J. Liu, A. Orbán, F. Güra, H. Zhou, X. Song, A. Boffy, H. Ganapathy, S. Zheng, H. Choe, Ágoston Weisz, T. Zhu, Y. Lu, S. Gopal, J. Kahn, M. Kula, J. Pitman, R. Shah, E. Taropa, M. A. Merey, M. Baeuml, Z. Chen, L. E. Shafey, Y. Zhang, O. Sercinoglu, G. Tucker, E. Piqueras, M. Krikun, I. Barr, N. Savinov, I. Danihelka, B. Roelofs, A. White, A. Andreassen, T. von Glehn, L. Yagati, M. Kazemi, L. Gonzalez, M. Khalman, J. Sygnowski, A. Fréchette, C. G. Smith, L. Culp, L. Proleev, Y. Luan, and X. Chen. Gemini: A family of highly capable multimodal models. *arXiv (Cornell University)*, 2023. doi: 10.48550/arxiv.2312.11805. URL <https://arxiv.org/pdf/2312.11805>.
- R. Thoppilan, D. D. Freitas, J. Hall, N. Shazeer, A. Kulshreshtha, H.-T. Cheng, A. Jin, T. Bos, L. L. Baker, Y. Du, Y. Li, H. Lee, H. Zheng, A. Ghafouri, M. Menegali, Y. Huang, M. Krikun, D. Lepikhin, J. Qin, D. Chen, Y. Xu, Z. Chen, A. Roberts, M. Bosma, V. Zhao, Y. Zhou, C.-C. Chang, I. Krivokon, W. Rusch, M. Pickett, P. Srinivasan, L. Man, K. Meier-Hellstern, M. R. Morris, T. Doshi, R. D. Santos, T. Duke, J. H. Søraaker, B. Zevenbergen, V. Prabhakaran, M. Díaz, B. Hutchinson, K. Olson, A. Molina, E. Hoffman-John, J. Lee, L. Aroyo, R. Rajakumar, A. Butryna, M. S. Lamm, V. Kuzmina, J. Fenton, A. J. Cohen, R. Bernstein, R. Kurzweil, B. Aguera-Arcas, C. Cui, M. Croak, E. H., and Q. V. Le. Lamda: Language models for dialog applications. *arXiv (Cornell University)*, 2022. doi: 10.48550/arxiv.2201.08239. URL <https://arxiv.org/pdf/2201.08239>.
- H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, Édouard Grave, and G. Lample. Llama: Open and efficient foundation language models. *arXiv (Cornell University)*, 2023. doi: 10.48550/arxiv.2302.13971. URL <https://arxiv.org/pdf/2302.13971>.
- A. Wang, A. Singh, J. Michael, F. Hill, O. Levy, and S. Bowman. Glue: A multi-task benchmark and analysis platform for natural language understanding. *Unknown*, 2018. doi: 10.18653/v1/w18-5446. URL <https://www.aclweb.org/anthology/W18-5446.pdf>.
- A. Wang, Y. Pruksachatkun, N. Nangia, A. Singh, J. Michael, F. Hill, O. Levy, and S. R. Bowman. Superglue: A stickier benchmark for general-purpose language understanding systems. *arXiv (Cornell University)*, 2019. doi: 10.48550/arxiv.1905.00537. URL <https://arxiv.org/pdf/1905.00537>.
- L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei, and J.-R. Wen. A survey on large language model based autonomous agents. *Frontiers of Computer Science*, 2024. doi: 10.1007/s11704-024-40231-1. URL <https://link.springer.com/content/pdf/10.1007/s11704-024-40231-1.pdf>.

- X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H., S. Narang, A. Chowdhery, and D. Zhou. Self-consistency improves chain of thought reasoning in language models. *arXiv (Cornell University)*, 2022. doi: 10.48550/arxiv.2203.11171. URL <https://arxiv.org/pdf/2203.11171>.
- Z. Zhang, X. Han, Z. Liu, X. Jiang, M. Sun, and Q. Liu. Ernie: Enhanced language representation with informative entities. *Unknown*, 2019. doi: 10.18653/v1/p19-1139. URL <https://www.aclweb.org/anthology/P19-1139.pdf>.
- W. X. Zhao, K. Zhou, J. Li, T. Tang, X. Wang, Y. Hou, Y. Min, B. Zhang, J. Zhang, Z. Dong, Y. Du, Y. Chen, Y. Chen, Z. Chen, J. Jiang, R. Ren, Y. Li, X. Tang, Z. Liu, P. Liu, J. Nie, J.-R. Wen, and J.-R. Wen. A survey of large language models. *Frontiers of Computer Science*, 2026. doi: 10.1007/s11704-026-60308-3. URL <https://link.springer.com/content/pdf/10.1007/s11704-026-60308-3.pdf>.
- L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, L. Zi, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. *arXiv (Cornell University)*, 2023. doi: 10.48550/arxiv.2306.05685. URL <https://arxiv.org/pdf/2306.05685>.